Project: Investigate a TMDB-MOVIES Dataset

Table of Contents

- Introduction
- Objective
- Questions
- Data Wrangling
- Exploratory Data Analysis
- Conclusions

Introduction

> This data set contains information about 10,000 movies collected from The Movie Database (TMDb), including user ratings and revenue. Certain columns, like 'cast' and 'genres', contain multiple values separated by pipe (|) characters. There are some odd characters in the 'cast' column. Don't worry about cleaning them. You can leave them as is. The final two columns ending with "_adj" show the budget and revenue of the associated movie in terms of 2010 dollars, accounting for inflation over time.

Objective

> As a moviegoer, I am not only getting hocked by the story line, cinematography and cast performance of a movie but also I would like to pay more attention to define a successful movie in terms of commercial aspect. Because there will be no one want to invest in a film production without getting a clear picture of how profitable it would be as launching them out aka Return of investment (ROI). Based on that, we may find features that can partially relate to its economic success in this dataset. The final objective is finding aspects to make the next blockbuster movie which is substantially profitable.

Questions

> Q1. How the numerical variables distribution are in the dataset? Make this help to lay the foundation of understanding of the data we have.
>
> Q2. How many movies has been commercially successful in this dataset?
>
> Q3. Which are top 10 directors who frequently directed successful movies?

> Q4. Which are top 10 movies in term of revenue?
>
> Q5. How is typically the runtime of successful movies?
>
> Q6. Which year does have the highest number of successful movies release?
>
> Q7. What are the top frequent genres were made? And How is the financial performance of the genres?
>
> Q8. What are the top 10 frequent casts? And How is the financial performance of them?
>
> Q9. How do the vote average score and vote count relate to the revenue of the movies?
>
> Q10. What are words most used in tagline?

In [ ]:
```
# Import packages that plan to use
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
%matplotlib inline
import seaborn as sns
```

Data Wrangling

In [ ]:
```
# Load the dataset
df_raw = pd.read_csv('tmdb-movies.csv')
df_raw.head()
```

Out[ ]:

	id	imdb_id	popularity	budget	revenue	original_title	cast	
0	135397	tt0369610	32.985763	150000000	1513528810	Jurassic World	Chris Pratt|Bryce Dallas Howard|Irrfan Khan|Vi...	
1	76341	tt1392190	28.419936	150000000	378436354	Mad Max: Fury Road	Tom Hardy|Charlize Theron|Hugh Keays-Byrne|Nic...	
2	262500	tt2908446	13.112507	110000000	295238201	Insurgent	Shailene Woodley|Theo James|Kate Winslet|Ansel...	http://www.thed

	id	imdb_id	popularity	budget	revenue	original_title	cast	
3	140607	tt2488496	11.173104	200000000	2068178225	Star Wars: The Force Awakens	Harrison Ford|Mark Hamill|Carrie Fisher|Adam D...	http://ww
4	168259	tt2820852	9.335014	190000000	1506249360	Furious 7	Vin Diesel|Paul Walker|Jason Statham|Michelle ...	

5 rows × 21 columns

In [ ]:
```
df_raw.tail()
```

Out[ ]:

	id	imdb_id	popularity	budget	revenue	original_title	cast	homepage	
10861	21	tt0060371	0.080598	0	0	The Endless Summer	Michael Hynson|Robert August|Lord 'Tally Ho' B...	NaN	
10862	20379	tt0060472	0.065543	0	0	Grand Prix	James Garner|Eva Marie Saint|Yves Montand|Tosh...	NaN	Fr
10863	39768	tt0060161	0.065141	0	0	Beregis Avtomobilya	Innokentiy Smoktunovskiy|Oleg Efremov|Georgi Z...	NaN	
10864	21449	tt0061177	0.064317	0	0	What's Up, Tiger Lily?	Tatsuya Mihashi|Akiko Wakabayashi|Mie Hama|Joh...	NaN	
10865	22293	tt0060666	0.035919	19000	0	Manos: The Hands of Fate	Harold P. Warren|Tom Neyman|John Reynolds|Dian...	NaN	

5 rows × 21 columns

In [ ]:
```
# Check how many columns, rows of the dataset are.
print(f" There are {df_raw.shape[0]} rows in the dataset.")
print(f" There are {df_raw.shape[1]} columns in the dataset.")
```

```
There are 10866 rows in the dataset.
There are 21 columns in the dataset.
```

In [ ]:
```
# Check the data type and information of the dataset
df_raw.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10866 entries, 0 to 10865
Data columns (total 21 columns):
 #   Column                Non-Null Count  Dtype  
---  ------                --------------  -----  
 0   id                    10866 non-null  int64  
 1   imdb_id               10856 non-null  object 
 2   popularity            10866 non-null  float64
 3   budget                10866 non-null  int64  
 4   revenue               10866 non-null  int64  
 5   original_title        10866 non-null  object 
 6   cast                  10790 non-null  object 
 7   homepage              2936 non-null   object 
 8   director              10822 non-null  object 
 9   tagline               8042 non-null   object 
 10  keywords              9373 non-null   object 
 11  overview              10862 non-null  object 
 12  runtime               10866 non-null  int64  
 13  genres                10843 non-null  object 
 14  production_companies  9836 non-null   object 
 15  release_date          10866 non-null  object 
 16  vote_count            10866 non-null  int64  
 17  vote_average          10866 non-null  float64
 18  release_year          10866 non-null  int64  
 19  budget_adj            10866 non-null  float64
 20  revenue_adj           10866 non-null  float64
dtypes: float64(4), int64(6), object(11)
memory usage: 1.7+ MB
```

- Following columns will be dropped due to not being in the circle of the analysis : id, imdb_id, hompage, keywords, overview, budget_adj, revenue_adj, popularity, production_companies, release_date

In [ ]:
```
# Dropout columns not being used in the analysis
df_raw.drop(columns=['id', 'imdb_id', 'homepage', 'keywords', 'overview', 'budget_adj',
df_raw.head()
```

Out[ ]:

	budget	revenue	original_title	cast	director	tagline	runtime	
0	150000000	1513528810	Jurassic World	Chris Pratt|Bryce Dallas Howard|Irrfan Khan|Vi...	Colin Trevorrow	The park is open.	124	Action|Adventu Fict

	budget	revenue	original_title	cast	director	tagline	runtime	
1	150000000	378436354	Mad Max: Fury Road	Tom Hardy|Charlize Theron|Hugh Keays-Byrne|Nic...	George Miller	What a Lovely Day.	120	Action|Adventu Fict
2	110000000	295238201	Insurgent	Shailene Woodley|Theo James|Kate Winslet|Ansel...	Robert Schwentke	One Choice Can Destroy You	119	Adventu Fict
3	200000000	2068178225	Star Wars: The Force Awakens	Harrison Ford|Mark Hamill|Carrie Fisher|Adam D...	J.J. Abrams	Every generation has a story.	136	Action|Adventu Ficti
4	190000000	1506249360	Furious 7	Vin Diesel|Paul Walker|Jason Statham|Michelle ...	James Wan	Vengeance Hits Home	137	Action|Cri

In [ ]:
```
df_raw.tail()
```

Out[ ]:

	budget	revenue	original_title	cast	director	tagline	runtime	
10861	0	0	The Endless Summer	Michael Hynson|Robert August|Lord 'Tally Ho' B...	Bruce Brown	NaN	95	
10862	0	0	Grand Prix	James Garner|Eva Marie Saint|Yves Montand|Tosh...	John Frankenheimer	Cinerama sweeps YOU into a drama of speed and ...	176	Action|
10863	0	0	Beregis Avtomobilya	Innokentiy Smoktunovskiy|Oleg Efremov|Georgi Z...	Eldar Ryazanov	NaN	94	
10864	0	0	What's Up, Tiger Lily?	Tatsuya Mihashi|Akiko Wakabayashi|Mie Hama|Joh...	Woody Allen	WOODY ALLEN STRIKES BACK!	80	
10865	19000	0	Manos: The Hands of Fate	Harold P. Warren|Tom Neyman|John Reynolds|Dian...	Harold P. Warren	It's Shocking! It's Beyond Your Imagination!	74	

In [ ]:
```
#Checking the descriptive statistics of numerical variables
df_raw.describe()
```

Out[ ]:

	budget	revenue	runtime	vote_count	vote_average	release_year
count	1.086600e+04	1.086600e+04	10866.000000	10866.000000	10866.000000	10866.000000
mean	1.462570e+07	3.982332e+07	102.070863	217.389748	5.974922	2001.322658
std	3.091321e+07	1.170035e+08	31.381405	575.619058	0.935142	12.812941
min	0.000000e+00	0.000000e+00	0.000000	10.000000	1.500000	1960.000000
25%	0.000000e+00	0.000000e+00	90.000000	17.000000	5.400000	1995.000000
50%	0.000000e+00	0.000000e+00	99.000000	38.000000	6.000000	2006.000000
75%	1.500000e+07	2.400000e+07	111.000000	145.750000	6.600000	2011.000000
max	4.250000e+08	2.781506e+09	900.000000	9767.000000	9.200000	2015.000000

- There are movies in which has zero budget, or zero revenue. These is not valuable at the purpose of analyzing commercial aspects of this report. Therefore, those values with zero commercial information will be droped out.
- Min runtime is zero which means the movie length was posibly input wrong. The maximum runtine in movie is 900 min which is subtatianlly long one. Let's have a look on that maxium 900 min movie.

In [ ]:
```
# Subset to movie where the runtime is 900min.
df_raw[df_raw['runtime'] == 900]
```

Out[ ]:

	budget	revenue	original_title	cast	director	tagline	runtime	genres	vote_coun
3894	0	0	The Story of Film: An Odyssey	Mark Cousins|Jean-Michel Frodon|Cari Beauchamp...	Mark Cousins	NaN	900	Documentary	1

The 900 min is "The Story of Film: An Odyssey" which is a documentary series with 15 episodes. Therefore, this movie is valid for the total cummulative of the length of 15 eps.

> https://en.wikipedia.org/wiki/The_Story_of_Film:_An_Odyssey#:~:text=The%20Story%20of%20Fil

In [ ]:
```
# Subset to movies where the runtime is 0min.
movie_0_runtime = df_raw[df_raw['runtime'] == 0]
movie_0_runtime
```

Out[ ]:

	budget	revenue	original_title	cast	director	tagline	runtime	
92	0	0	Mythica: The Necromancer	Melanie Stone|Adam Johnson|Kevin Sorbo|Nicola ...	A. Todd Smith	NaN	0	Fantasy

	budget	revenue	original_title	cast	director	tagline	runtime	
334	0	0	Ronaldo	Cristiano Ronaldo	Anthony Wonke	Astonishing. Intimate. Definitive.	0	
410	0	0	Anarchy Parlor	Robert LaSardo|Jordan James Smith|Sara Fabel|T...	Kenny Gage|Devon Downs	NaN	0	
445	0	0	The Exorcism of Molly Hartley	Sarah Lind|Devon Sawa|Gina Holden|Peter MacNei...	Steven R. Monroe	NaN	0	
486	0	0	If There Be Thorns	Heather Graham|Jason Lewis|Rachael Carpani|Mas...	Nancy Savoca	NaN	0	
595	0	0	Deep Dark	Sean McGrath|Denise Poirier|Anne Sorce|Tabor H...	Michael Medaglia	NaN	0	
616	0	0	The Outfield	Cameron Dallas|Hamilton Nash Grier|Olivia Stuc...	Michael Goldfine	NaN	0	
1241	0	0	Dance-Off	Kathryn McCormick|Shane Harper|Finola Hughes|C...	NaN	NaN	0	Roma
1289	3250000	0	Treehouse	J. Michael Trautmann|Dana Melanie|Daniel Fredr...	Michael G. Bartlett	No kids allowed	0	Thri
1293	0	0	Tim Maia	Robson Nunes|BabÃº Santana|Alinne Moraes|CauÃ£...	Mauro Lima	Original language PORTUGUESE	0	Docume
1849	0	0	Spectacular!	Nolan Gerard Funk|Jesse Moss|Britt Irvin|Tammi...	Robert Iscove	NaN	0	
2315	0	0	Listen to Your Heart	Alexia Rasmussen|Kent Moran|Cybill Shepherd|Sh...	NaN	One song can change your life.	0	Dra
2370	0	0	Freshman Father	Britt Irvin|Merrilyn Gann|Barbara Tyson|Anthon...	Michael Scott	NaN	0	

	budget	revenue	original_title	cast	director	tagline	runtime	
3329	0	0	Grande, grosso e Verdone	Carlo Verdone|Claudia Gerini|Eva Riccobono|Vit...	Carlo Verdone	NaN	0	
3794	0	0	Toi, moi, les autres	LeÃ¯la Bekhti|Benjamin Siksou|CÃ©cile Cassel|C...	Audrey Estrougo	NaN	0	
3857	0	0	Cell 213	Bruce Greenwood|Eric Balfour|Michael Rooker|De...	Stephen Kay	NaN	0	
3884	0	0	eCupid	Houston Rhines|Andy Anderson|John Callahan|Gal...	J.C. Calciano	NaN	0	
4063	6000000	0	Madea's Family Reunion	Tyler Perry|Blair Underwood|Lynn Whitfield|Bor...	Tyler Perry	NaN	0	
4138	0	0	A Time for Dancing	Larisa Oleynik|Shiri Appleby|Peter Coyote|Patr...	Peter Gilbert	NaN	0	
4829	0	0	Rags	Keke Palmer|Avan Jogia|Drake Bell|Burkely Duff...	Bille Woodruff	NaN	0	TV Mov
4944	0	0	How to Fall in Love	Brooke D'Orsay|Eric Mabius|Jody Thompson|Gina ...	Mark Griffiths	NaN	0	Comedy|R
5216	0	0	Madea's Class Reunion	Tyler Perry|Chandra Currelley|Cheryl Pepsii Ri...	Tyler Perry	NaN	0	
5695	0	0	Skinwalker Ranch	Steve Berg|Kyle Davis|Erin Cahill|Jon Gries|De...	Devin McGinn	NaN	0	Thri
5920	0	0	The Food Guide to Love	Richard Coyle|Leonor Watling|GinÃ©s GarcÃa Mi...	Dominic Harari|Teresa Pelegri	NaN	0	
5938	0	0	Go Goa Gone	Saif Ali Khan|Anand Tiwari|Vir Das|Pooja Gupta	Krishna D.K.|Raj Nidimoru	NaN	0	

	budget	revenue	original_title	cast	director	tagline	runtime	
5992	0	0	Amiche da morire	Claudia Gerini|Cristiana Capotondi|Sabrina Imp...	Giorgia Farina	NaN	0	Romai
6040	0	0	The Vatican Exorcisms	Joe Marino	Joe Marino	The public were never meant to know	0	Horror|Doc
6383	2000000	0	The 12 Dogs of Christmas	Jordan-Claire Green|Tom Kemp|Susan Wood|Adam H...	Kieth Merrill	NaN	0	
6552	0	0	Quatre Ã©toiles	JosÃ© Garcia|Isabelle CarrÃ©|RenÃ©e Le Calm|Fr...	Christian Vincent	NaN	0	
6934	0	0	Jean-Philippe	Fabrice Luchini|Johnny Hallyday|Jackie Berroye...	Laurent Tuel	NaN	0	
8874	0	0	Mission Kashmir	Sanjay Dutt|Hrithik Roshan|Preity Zinta|Jackie...	Vidhu Vinod Chopra	NaN	0	Act

In [ ]:
```
# Check how movies have 0 runtime
df_raw[df_raw['runtime'] == 0].shape[0]
```

Out[ ]: 31

There are 31 movies with the runtime is zero. And most of them has zero on revenue. That will not contribute into the analysis, therefore, I will drop them out of the dataset.

In [ ]:
```
# Drop rows with 0 runtime
df_raw.drop(movie_0_runtime.index, inplace=True)
```

In [ ]:
```
# Reset index of the dataframe
df_raw.reindex()
```

Out[ ]:

	budget	revenue	original_title	cast	director	tagline	runtime
0	150000000	1513528810	Jurassic World	Chris Pratt|Bryce Dallas Howard|Irrfan Khan|Vi...	Colin Trevorrow	The park is open.	124
1	150000000	378436354	Mad Max: Fury Road	Tom Hardy|Charlize Theron|Hugh Keays-Byrne|Nic...	George Miller	What a Lovely Day.	120

	budget	revenue	original_title	cast	director	tagline	runtime
2	110000000	295238201	Insurgent	Shailene Woodley|Theo James|Kate Winslet|Ansel...	Robert Schwentke	One Choice Can Destroy You	119
3	200000000	2068178225	Star Wars: The Force Awakens	Harrison Ford|Mark Hamill|Carrie Fisher|Adam D...	J.J. Abrams	Every generation has a story.	136
4	190000000	1506249360	Furious 7	Vin Diesel|Paul Walker|Jason Statham|Michelle ...	James Wan	Vengeance Hits Home	137
...	...	...	...	...	...	...	...
10861	0	0	The Endless Summer	Michael Hynson|Robert August|Lord 'Tally Ho' B...	Bruce Brown	NaN	95
10862	0	0	Grand Prix	James Garner|Eva Marie Saint|Yves Montand|Tosh...	John Frankenheimer	Cinerama sweeps YOU into a drama of speed and ...	176
10863	0	0	Beregis Avtomobilya	Innokentiy Smoktunovskiy|Oleg Efremov|Georgi Z...	Eldar Ryazanov	NaN	94
10864	0	0	What's Up, Tiger Lily?	Tatsuya Mihashi|Akiko Wakabayashi|Mie Hama|Joh...	Woody Allen	WOODY ALLEN STRIKES BACK!	80
10865	19000	0	Manos: The Hands of Fate	Harold P. Warren|Tom Neyman|John Reynolds|Dian...	Harold P. Warren	It's Shocking! It's Beyond Your Imagination!	74

10835 rows × 11 columns

In [ ]:
```
#Check the runtime 0 is gone
df_raw[df_raw['runtime'] == 0].shape[0]
```

Out[ ]:
```
0
```

In [ ]:
```
#Checking any null values in the dataset
df_raw.isnull().sum()
```

Out[ ]:
```
budget              0
revenue             0
original_title      0
cast               76
director           42
```

```
tagline           2798
runtime              0
genres              22
vote_count           0
vote_average         0
release_year         0
dtype: int64
```

Most of the null value was on categorical varibales: cast, director, tagline, genres, production companies. These are not so important for now to drop them out so I would to keep them for now.

In [ ]:
```
#Check the duplicated rows
df_raw.duplicated().sum()
```

Out[ ]:
```
1
```

In [ ]:
```
#Drop duplicated row
df_raw.drop_duplicates(inplace=True)
df_raw.reindex()
df_raw.duplicated().sum()
```

Out[ ]:
```
0
```

Drop out rows that has 0 on either revenue, budget column which is unable to analyse them on commercial aspect

In [ ]:
```
# Check how many movies have zero budget
movie_0_budget = df_raw.query('budget == 0')
movie_0_budget.shape[0]
```

Out[ ]:
```
5668
```

In [ ]:
```
# Double check all those 0 budget movie is gone
df_raw.drop(movie_0_budget.index, inplace=True)
df_raw.query('budget == 0').shape[0]
```

Out[ ]:
```
0
```

In [ ]:
```
# Do the same for those are 0 revenue
movie_0_revenue = df_raw.query('revenue == 0')
df_raw.drop(movie_0_revenue.index, inplace=True)
df_raw.query('budget == 0').shape[0]
```

Out[ ]:
```
0
```

In [ ]:
```
# Check again the dataset statistics
df_raw.describe()
```

Out[ ]:

	budget	revenue	runtime	vote_count	vote_average	release_year
count	3.854000e+03	3.854000e+03	3854.000000	3854.000000	3854.000000	3854.000000

	budget	revenue	runtime	vote_count	vote_average	release_year
mean	3.720370e+07	1.076866e+08	109.220291	527.720291	6.168163	2001.261028
std	4.220822e+07	1.765393e+08	19.922820	879.956821	0.794920	11.282575
min	1.000000e+00	2.000000e+00	15.000000	10.000000	2.200000	1960.000000
25%	1.000000e+07	1.360003e+07	95.000000	71.000000	5.700000	1995.000000
50%	2.400000e+07	4.480000e+07	106.000000	204.000000	6.200000	2004.000000
75%	5.000000e+07	1.242125e+08	119.000000	580.000000	6.700000	2010.000000
max	4.250000e+08	2.781506e+09	338.000000	9767.000000	8.400000	2015.000000

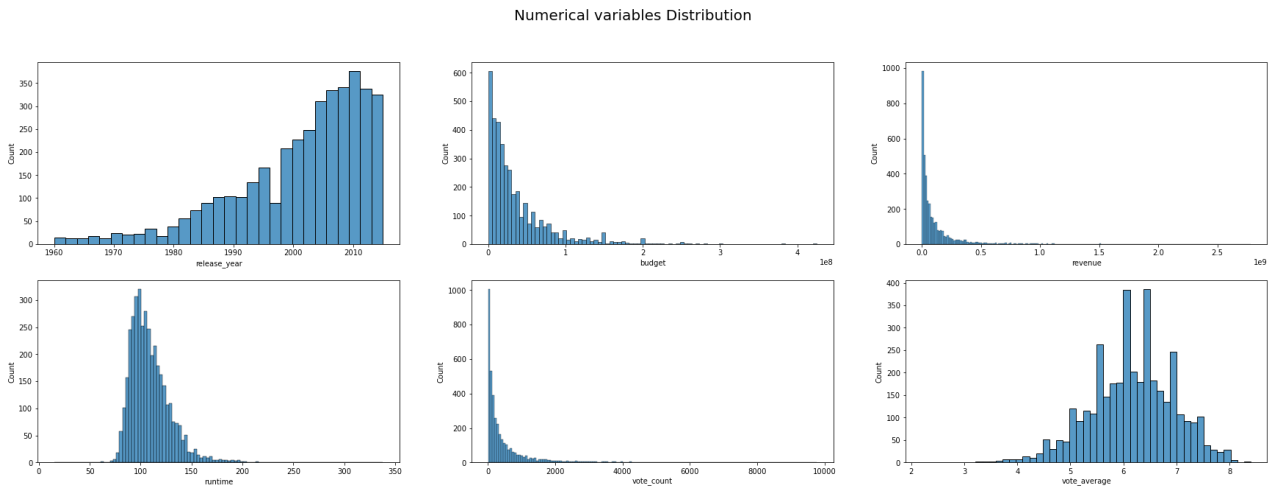
Exploratory Data Analysis

Q1. How the numerical variables distribution are in the dataset? Make this help to lay the foundation of understanding of the data we have.

In [ ]:

```
#Make the histogram plot for all numerical variables
fig, axes = plt.subplots(2, 3, figsize=(30, 10))
fig.suptitle('Numerical variables Distribution', size=20)

sns.histplot(ax=axes[0, 0], data=df_raw['release_year'])
sns.histplot(ax=axes[0, 1], data=df_raw['budget'])
sns.histplot(ax=axes[0, 2], data=df_raw['revenue'])
sns.histplot(ax=axes[1, 0], data=df_raw['runtime'])
sns.histplot(ax=axes[1, 1], data=df_raw['vote_count'])
sns.histplot(ax=axes[1, 2], data=df_raw['vote_average']);
```



- Release_year distribution: Number of movies is increasing rapidly after 20th century.
- Budget distribution: Most of the movie budget is below 100M. Highly skewed ro the right.
- Revenue distribution: Most of the movie revenue is below 500M. Heavily skewed ro the right.
- Runtime: lightly skewed to the right, most of the movie length is around 100min.
- Vote count: most of the movie got below 2000 votes. Heavily skewed ro the right.

- Vote_average: the distribution of the vote score is lightly on bell-shape curve. That's interesting.

Q2. How many movies has been commercially successful in this dataset?

In [ ]:
```
#Enrich the dataset by adding "profit" column = "budget" - "revenue"
df_raw['profit'] = df_raw['revenue'] - df_raw['budget']
#Enrich the dataset by adding "gross_margin" column = ("revenue"-"budget")/"budget". Th
df_raw['gross_margin'] = (df_raw['revenue'] - df_raw['budget'])/df_raw['budget']*100
df_raw.head()
```

Out[ ]:

	budget	revenue	original_title	cast	director	tagline	runtime	
0	150000000	1513528810	Jurassic World	Chris Pratt|Bryce Dallas Howard|Irrfan Khan|Vi...	Colin Trevorrow	The park is open.	124	Action|Adventu Fict
1	150000000	378436354	Mad Max: Fury Road	Tom Hardy|Charlize Theron|Hugh Keays-Byrne|Nic...	George Miller	What a Lovely Day.	120	Action|Adventu Fict
2	110000000	295238201	Insurgent	Shailene Woodley|Theo James|Kate Winslet|Ansel...	Robert Schwentke	One Choice Can Destroy You	119	Adventu Fict
3	200000000	2068178225	Star Wars: The Force Awakens	Harrison Ford|Mark Hamill|Carrie Fisher|Adam D...	J.J. Abrams	Every generation has a story.	136	Action|Adventu Ficti
4	190000000	1506249360	Furious 7	Vin Diesel|Paul Walker|Jason Statham|Michelle ...	James Wan	Vengeance Hits Home	137	Action|Cri

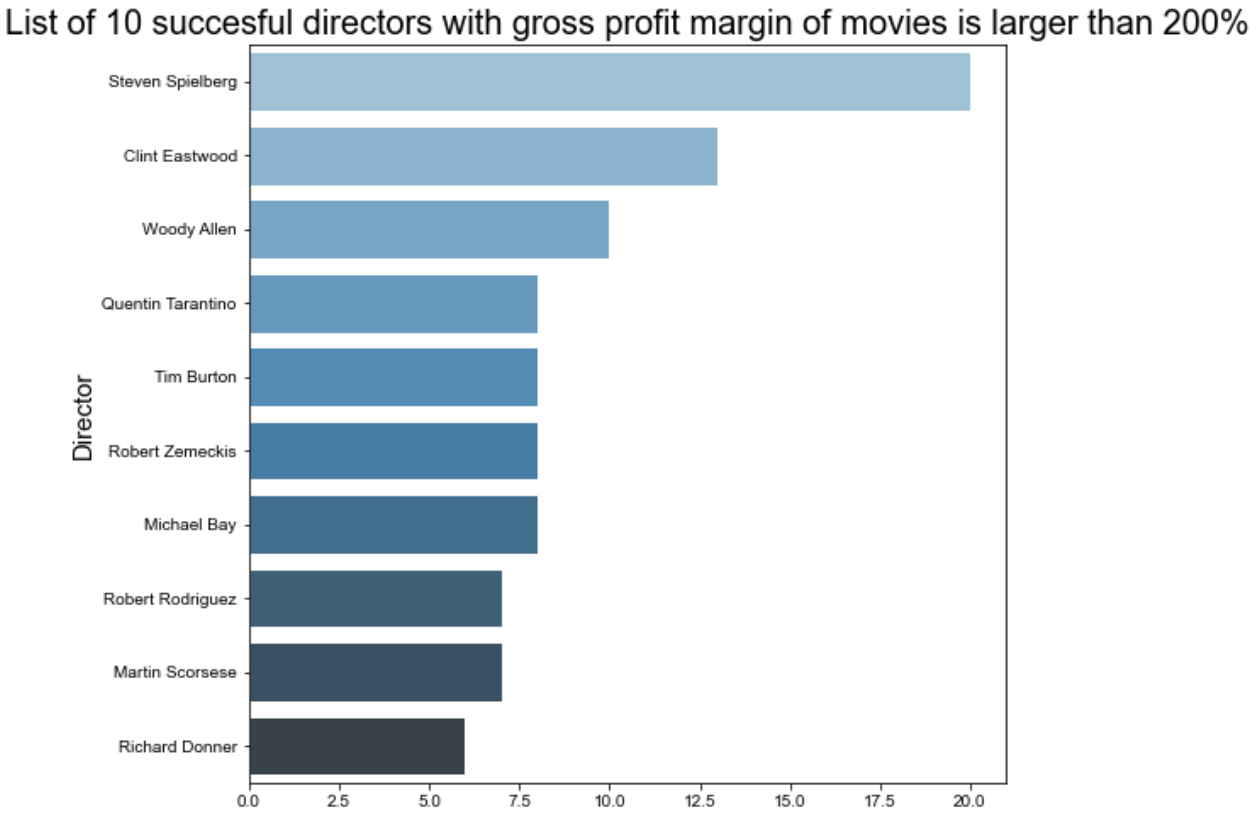
- A successful movie in term of commerciality, the gross margin is approximately 200% which means got double from the investment (budget). Let's have a look which movies on this top-notch list.

In [ ]:
```
df_commercial_success = df_raw.query('gross_margin >=200')
df_commercial_success.head
print(f" There are {df_commercial_success.shape[0]} movies has the profit margin larger
```

```
 There are 1405 movies has the profit margin larger than 200% and it takes 36.4556305137
5195 percent of the whole dataset.
```

Q3. Which are top 10 directors who frequently directed successful movies?

In [ ]:
```
#Top 10 sucessfull directors in term of commerciality.
success_director = df_commercial_success.groupby('director')['original_title'].count().
fig, axes = plt.subplots(figsize=(8, 8))
sns.set_style('darkgrid')
sns.barplot(y=success_director[:10].index, x=success_director[:10].values, palette='Blu
axes.set_title("List of 10 succesful directors with gross profit margin of movies is la
axes.set_ylabel("Director", size=15);
```

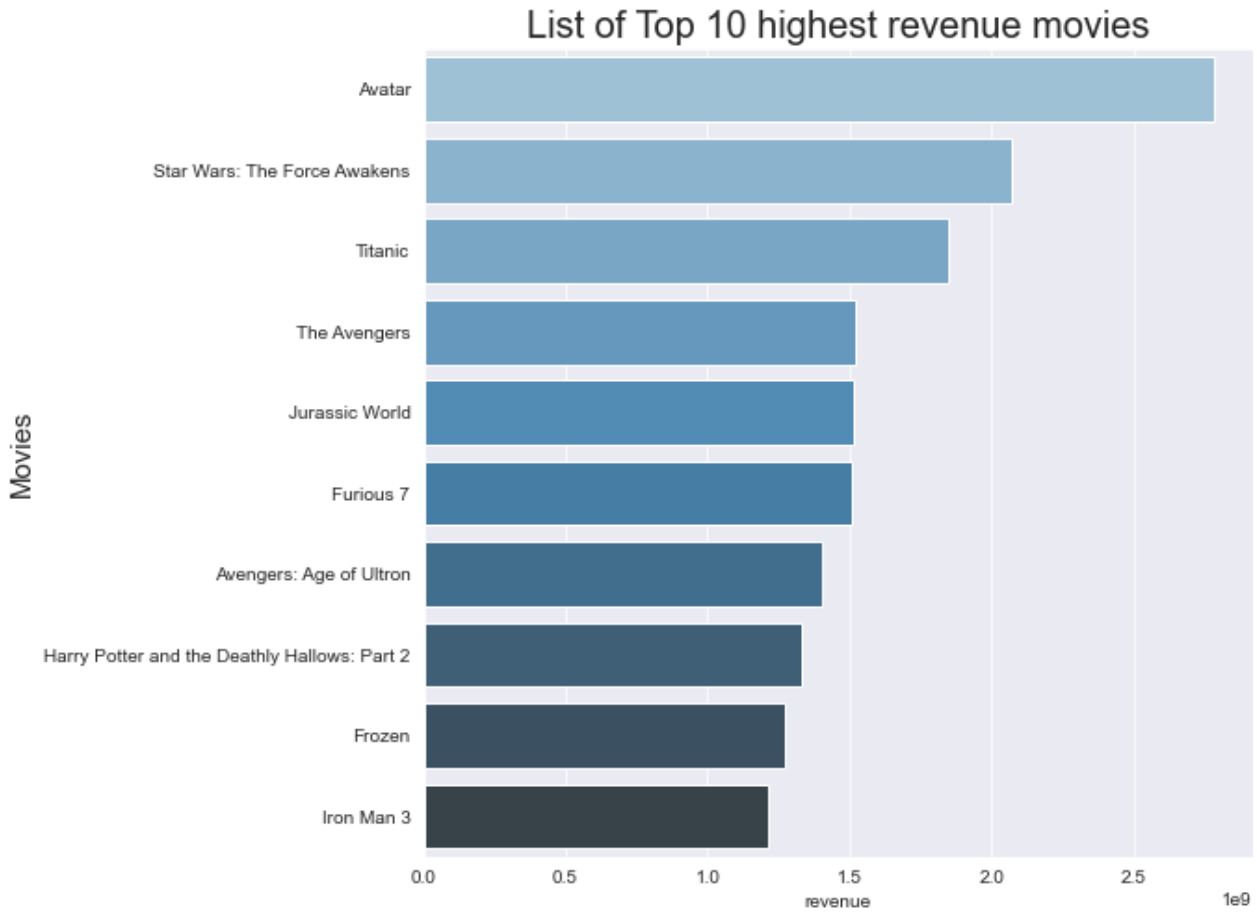


- If I want to make a next profitable blockbuster movie, these directors are absolutely in my list of choice. They have directed a lot of movies in which the ROI is tremendous successful.

Q4. Which are top 10 movies in term of revenue?

In [ ]:
```
#Top 10 sucessfull movies in term of commerciality.
df_sorted_by_revenue = df_commercial_success.sort_values('revenue', ascending=False)

#Make the bar plot in seaborn to show top 10 movies in revenue.
fig, axes = plt.subplots(figsize=(8, 8))
sns.set_style('darkgrid')
sns.barplot(y=df_sorted_by_revenue['original_title'][:10], x=df_sorted_by_revenue['reve
axes.set_title("List of Top 10 highest revenue movies", size=20),
axes.set_ylabel("Movies", size=15);
```

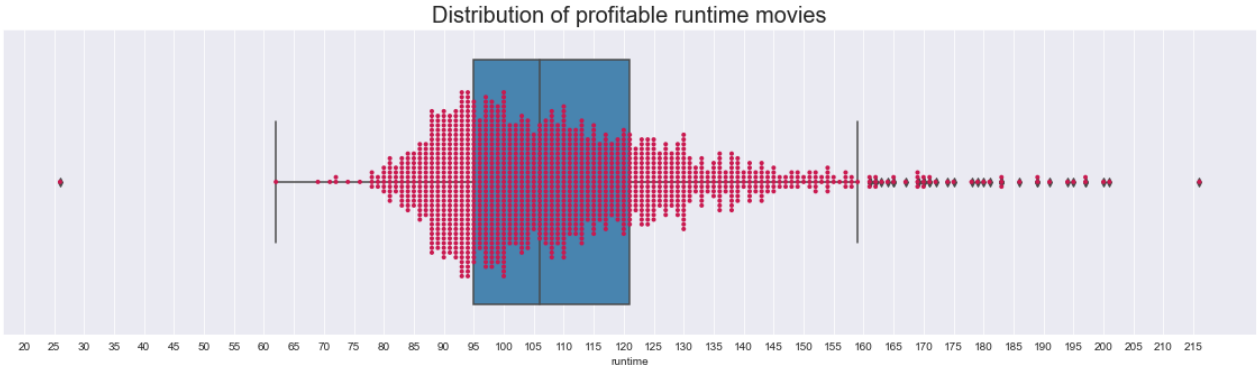


- All these blockbusters of the Hollywood have made more than 1B revenue in box-office. That's a heap of money.

Q5. How is typically the runtime of successful movies?

In [ ]:

```
# As we have been awared of this dataset, most of the movie length is around 100min. Le
fig, axes = plt.subplots(figsize=(20, 5))
axes = sns.boxplot(x='runtime', data=df_commercial_success, palette='Blues_d')
axes = sns.swarmplot(x='runtime', data=df_commercial_success, palette='rocket_r', color
axes.set_xticks(np.arange(20,220,5))
axes.set_title("Distribution of profitable runtime movies", size=20);
```

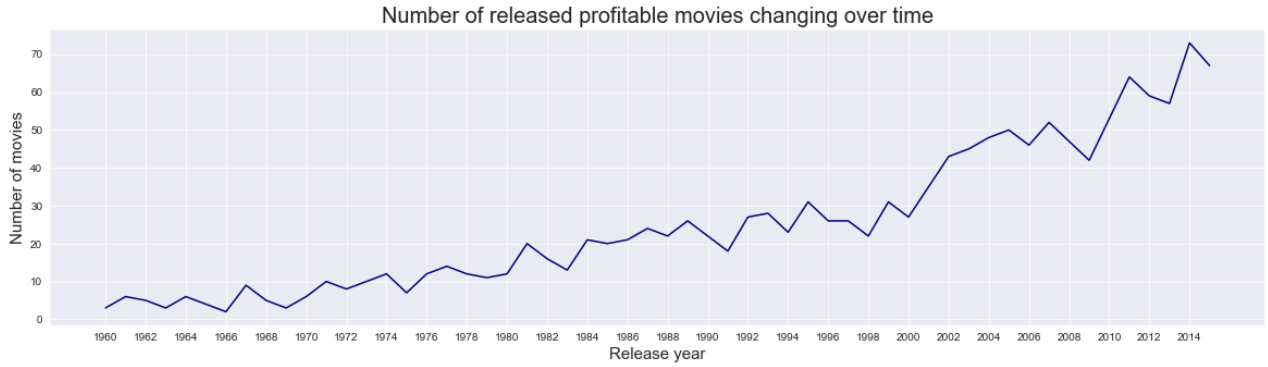


- 75% sucessful movie is below 122min and the highest numbers of movies is on 94min. Therefore, I would consider the profitable one in within this range.

- I have asked a friend of mine who is a financial cinebox's chain manager here in Ho Chi Minh City, Vietnam. He said that this range of runtime 94-122min is one the of best time frame to maximize the number of schedules. That's reason why they will generate a big amount of money in the box-office.

Q6. Which year does have the highest number of successful movies release?

In [ ]:
```
# Group by release year and count how many movies is changing over time.
fig, axes = plt.subplots(figsize=(20, 5))
df_commercial_success.groupby('release_year')['original_title'].count().plot(kind='line
axes.set_xticks(np.arange(1960,2015,2))
axes.set_title("Number of released profitable movies changing over time", size=20)
axes.set_ylabel("Number of movies", size=15)
axes.set_xlabel("Release year", size=15);
```



- The movie industry is getting more profitable movies over time on overall trend. The peak year is in 2014 in which there are 70 commercial success movies were made. That indicates the movie industry will be a promising field to get the investment in the future.

Q7. What are the top frequent genres were made? And How is the financial performance of the genres?

In [ ]:
```
# split genres string, at |, explode the list to rows
genres_df = df_commercial_success.assign(genres=df_commercial_success['genres'].str.spl
genres_df.head(10)
```

Out[ ]:

	budget	revenue	original_title	cast	director	tagline	runtime	genres	vo
0	150000000	1513528810	Jurassic World	Chris Pratt|Bryce Dallas Howard|Irrfan Khan|Vi...	Colin Trevorrow	The park is open.	124	Action	
0	150000000	1513528810	Jurassic World	Chris Pratt|Bryce Dallas Howard|Irrfan Khan|Vi...	Colin Trevorrow	The park is open.	124	Adventure	

	budget	revenue	original_title	cast	director	tagline	runtime	genres	vo
0	150000000	1513528810	Jurassic World	Chris Pratt|Bryce Dallas Howard|Irrfan Khan|Vi...	Colin Trevorrow	The park is open.	124	Science Fiction	
0	150000000	1513528810	Jurassic World	Chris Pratt|Bryce Dallas Howard|Irrfan Khan|Vi...	Colin Trevorrow	The park is open.	124	Thriller	
3	200000000	2068178225	Star Wars: The Force Awakens	Harrison Ford|Mark Hamill|Carrie Fisher|Adam D...	J.J. Abrams	Every generation has a story.	136	Action	
3	200000000	2068178225	Star Wars: The Force Awakens	Harrison Ford|Mark Hamill|Carrie Fisher|Adam D...	J.J. Abrams	Every generation has a story.	136	Adventure	
3	200000000	2068178225	Star Wars: The Force Awakens	Harrison Ford|Mark Hamill|Carrie Fisher|Adam D...	J.J. Abrams	Every generation has a story.	136	Science Fiction	
3	200000000	2068178225	Star Wars: The Force Awakens	Harrison Ford|Mark Hamill|Carrie Fisher|Adam D...	J.J. Abrams	Every generation has a story.	136	Fantasy	
4	190000000	1506249360	Furious 7	Vin Diesel|Paul Walker|Jason Statham|Michelle ...	James Wan	Vengeance Hits Home	137	Action	
4	190000000	1506249360	Furious 7	Vin Diesel|Paul Walker|Jason Statham|Michelle ...	James Wan	Vengeance Hits Home	137	Crime	

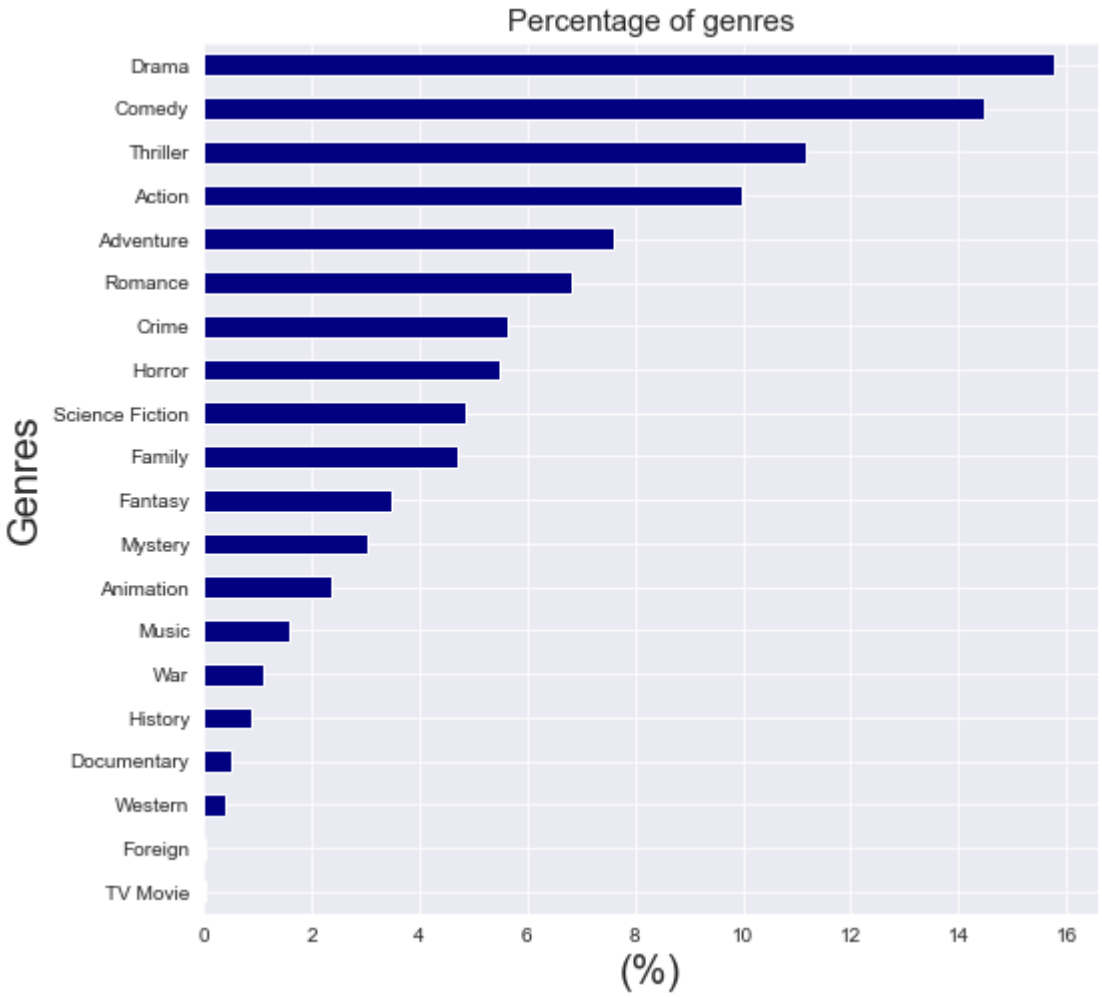
In [ ]:
```
#Which genres is most directed?
genres_pct_count = genres_df['genres'].value_counts(normalize=True)*100
genres_pct_count
```

Out[ ]:
```
Drama              15.780731
Comedy             14.451827
Thriller           11.157254
Action              9.994463
Adventure           7.613511
Romance             6.838317
Crime               5.647841
Horror              5.481728
Science Fiction     4.844961
Family              4.706534
Fantasy             3.488372
Mystery             3.045404
Animation           2.353267
Music               1.605759
```

```
War                1.107420
History            0.885936
Documentary        0.526024
Western            0.415282
Foreign            0.027685
TV Movie           0.027685
Name: genres, dtype: float64
```

In [ ]:

```
# Plot the barchart of the percentage of movie genres appeared
fig, axes = plt.subplots(figsize=(8, 8))
genres_pct_count.sort_values().plot(kind='barh', color="navy")
axes.set_title("Percentage of genres", size=15)
axes.set_ylabel("Genres", size=20)
axes.set_xlabel("(%)", size=20);
```



- Drama (15.78%), Comedy (14.45%), Thriller (11.15%) are the top 3 genres are being made and take 41.38% in total number.

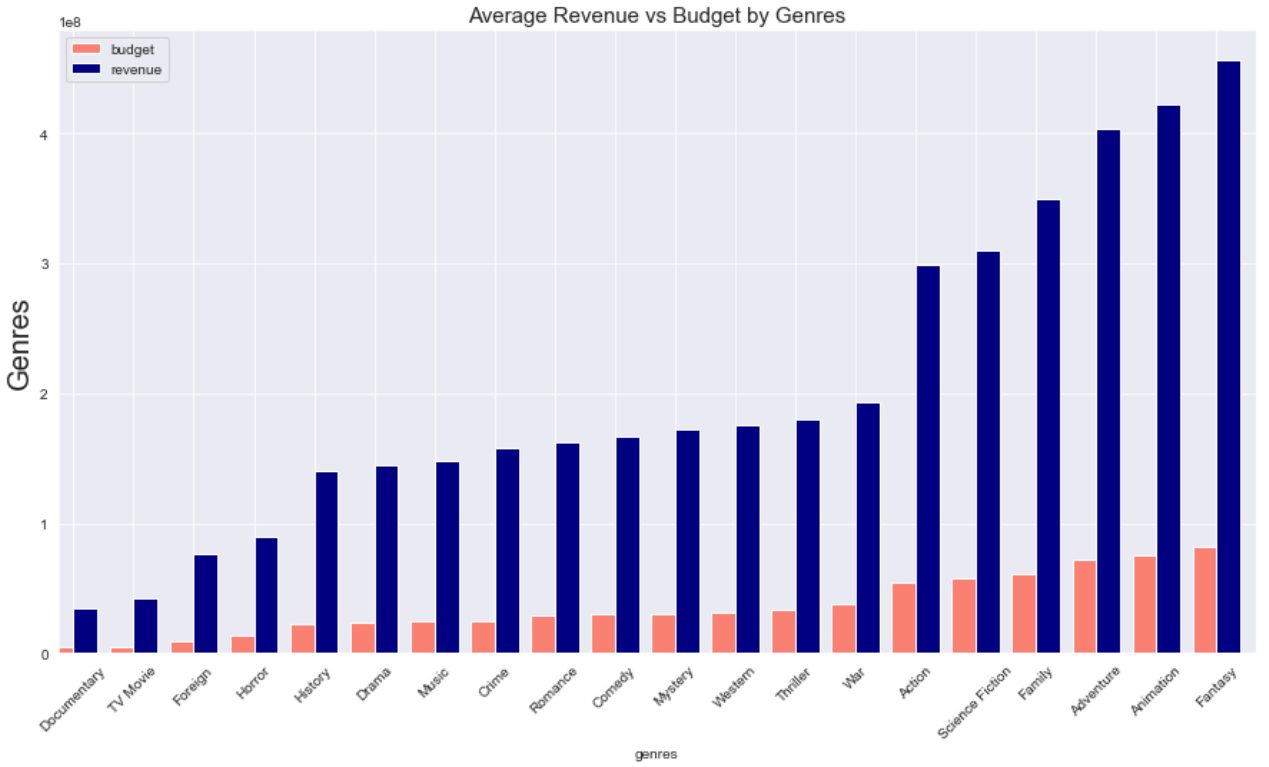
In [ ]:

```
#Calculate the average budget/revenue of each genres
avg_genres_revenue = genres_df.groupby('genres')['revenue'].mean().sort_values(ascendin
avg_genres_budget = genres_df.groupby('genres')['budget'].mean().sort_values(ascending=

#Plot double bar chart for both average values
fig = plt.figure(figsize=(15,8))
axes = fig.add_subplot(111)
width = 0.4
```

```
avg_genres_budget.plot(kind='bar', color='salmon', ax=axes, width=width, position=1)
avg_genres_revenue.plot(kind='bar', color='navy', ax=axes, width=width, position=0)

axes.set_title("Average Revenue vs Budget by Genres", size=15)
axes.set_ylabel("Genres", size=20)
plt.xticks(rotation = 45)
plt.legend();
```



- The bar chart clearly show that the more budget the more revenue in return. Fantasy has the largest average revenue (almost 500M); however, this genre also consumes alot of the budget to make it.
- Interestingly, my pre-perception is that animation movie has no cast-acting involved in the movie therefore the budget could be lower than others. However, the budget for this one stay on the second place but still can generate a big amount of revenue (more than 400M).
- As we know drama, comedy & thriller are the top frequent genres made, but the revenue of them is not of the top 3 in term of revenue and budget.

Q8. What are the top 10 frequent casts? And How is the financial performance of them?

In [ ]:
```
# split cast string, at |, explode the list to rows
cast_df = df_commercial_success.assign(cast=df_commercial_success['cast'].str.split('|'
cast_df.head(10)
```

Out[ ]:

	budget	revenue	original_title	cast	director	tagline	runtime	genr
0	150000000	1513528810	Jurassic World	Chris Pratt	Colin Trevorrow	The park is open.	124	Action|Adventure|Scien Fiction|Thril

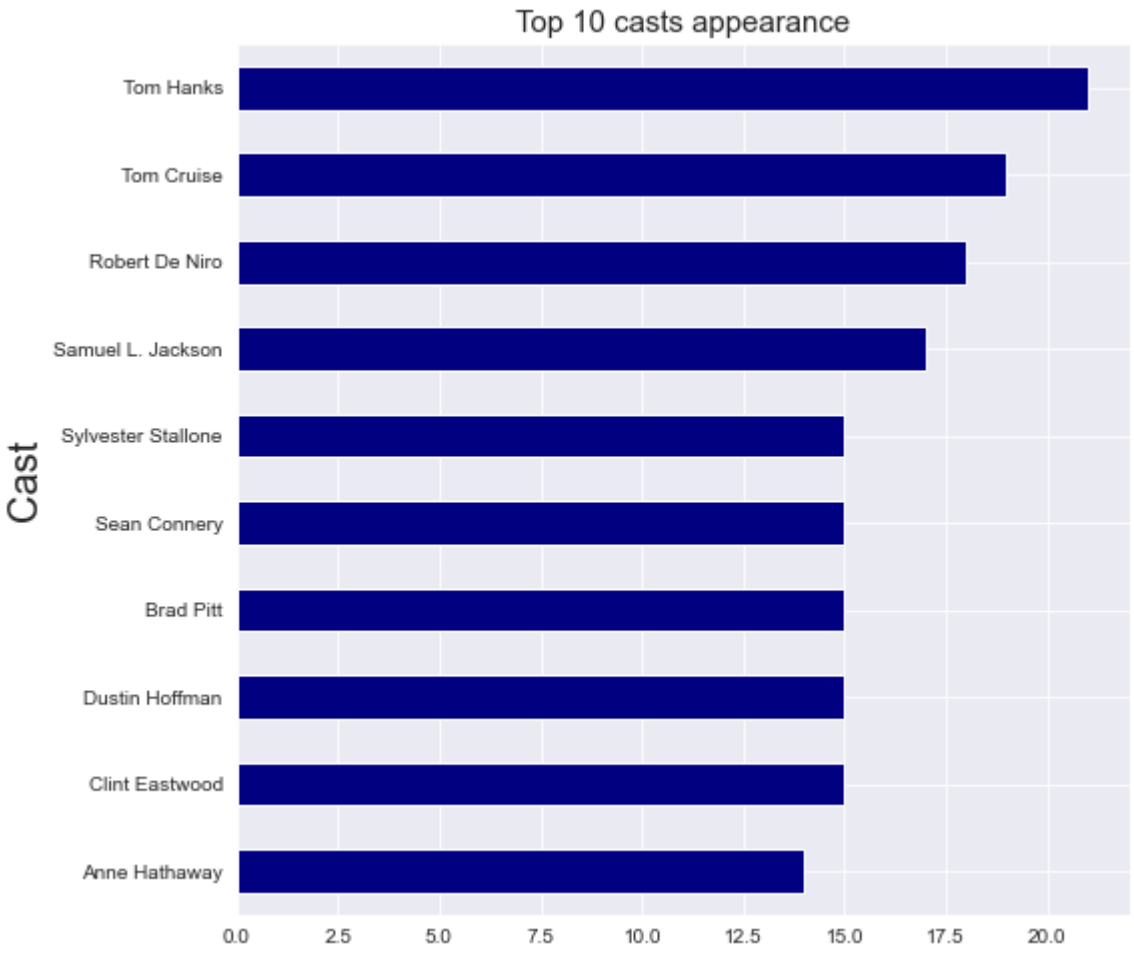
	budget	revenue	original_title	cast	director	tagline	runtime	genr
0	150000000	1513528810	Jurassic World	Bryce Dallas Howard	Colin Trevorrow	The park is open.	124	Action|Adventure|Scien Fiction|Thril
0	150000000	1513528810	Jurassic World	Irrfan Khan	Colin Trevorrow	The park is open.	124	Action|Adventure|Scien Fiction|Thril
0	150000000	1513528810	Jurassic World	Vincent D'Onofrio	Colin Trevorrow	The park is open.	124	Action|Adventure|Scien Fiction|Thril
0	150000000	1513528810	Jurassic World	Nick Robinson	Colin Trevorrow	The park is open.	124	Action|Adventure|Scien Fiction|Thril
3	200000000	2068178225	Star Wars: The Force Awakens	Harrison Ford	J.J. Abrams	Every generation has a story.	136	Action|Adventure|Scien Fiction|Fanta
3	200000000	2068178225	Star Wars: The Force Awakens	Mark Hamill	J.J. Abrams	Every generation has a story.	136	Action|Adventure|Scien Fiction|Fanta
3	200000000	2068178225	Star Wars: The Force Awakens	Carrie Fisher	J.J. Abrams	Every generation has a story.	136	Action|Adventure|Scien Fiction|Fanta
3	200000000	2068178225	Star Wars: The Force Awakens	Adam Driver	J.J. Abrams	Every generation has a story.	136	Action|Adventure|Scien Fiction|Fanta
3	200000000	2068178225	Star Wars: The Force Awakens	Daisy Ridley	J.J. Abrams	Every generation has a story.	136	Action|Adventure|Scien Fiction|Fanta

In [ ]:
```
# Top 10 frequent casts
cast_df['cast'].value_counts()
```

Out[ ]:
```
Tom Hanks            21
Tom Cruise           19
Robert De Niro       18
Samuel L. Jackson    17
Clint Eastwood       15
                     ..
Steve Huison          1
Eugene Byrd           1
Ben Mendelsohn        1
Esther Wurmfeld       1
JoJo Kushner          1
Name: cast, Length: 3790, dtype: int64
```

In [ ]:
```
#Plot the top 10 frequent casts
fig, axes = plt.subplots(figsize=(8, 8))
cast_count = cast_df['cast'].value_counts()
cast_count.iloc[:10].sort_values(ascending=True).plot(kind='barh', color="navy")
```

```
axes.set_title("Top 10 casts appearance", size=15)
axes.set_ylabel("Cast", size=20);
```



In [ ]:
```
# Convert the series to the list to subset them in the dataframe
most_demand_cast = cast_count.index[:10]
list(most_demand_cast)
```

Out[ ]:
```
['Tom Hanks',
 'Tom Cruise',
 'Robert De Niro',
 'Samuel L. Jackson',
 'Clint Eastwood',
 'Dustin Hoffman',
 'Brad Pitt',
 'Sean Connery',
 'Sylvester Stallone',
 'Anne Hathaway']
```

In [ ]:
```
### another way to subset rows with a series
selected_cast = list(most_demand_cast)
ten_most_demend_cast_query = cast_df.query('cast in @selected_cast')
ten_most_demend_cast_query
```

Out[ ]:

	budget	revenue	original_title	cast	director	tagline	runtime

	budget	revenue	original_title	cast	director	tagline	runtime	
15	44000000	155760117	The Hateful Eight	Samuel L. Jackson	Quentin Tarantino	No one comes up here without a damn good reason.	167	Crime|Drama|My
24	28000000	133346506	The Big Short	Brad Pitt	Adam McKay	This is a true story.	130	C
25	150000000	682330139	Mission: Impossible - Rogue Nation	Tom Cruise	Christopher McQuarrie	Desperate Times. Desperate Measures.	131	
27	81000000	403802136	Kingsman: The Secret Service	Samuel L. Jackson	Matthew Vaughn	Manners maketh man.	130	Crime|Comedy|Act
33	40000000	162610473	Bridge of Spies	Tom Hanks	Steven Spielberg	In the shadow of war, one man showed the world...	141	
...	...	...	...	...	...	...	...	
10596	500000	3000000	Mean Streets	Robert De Niro	Martin Scorsese	You don't make up for your sins in church. You...	110	
10608	12000000	53267000	Papillon	Dustin Hoffman	Franklin J. Schaffner	Not just a Great Escape. It was the Greatest.	151	
10689	11000000	141195658	Thunderball	Sean Connery	Terence Young	Look up! Look down! Look out!	130	Adventure|
10727	3600000	44785053	Midnight Cowboy	Dustin Hoffman	John Schlesinger	Whatever you hear about Midnight Cowboy is true.	113	
10762	15000000	50000000	The Deer Hunter	Robert De Niro	Michael Cimino	One of the most important and powerful films o...	183	

164 rows × 13 columns

In [ ]:

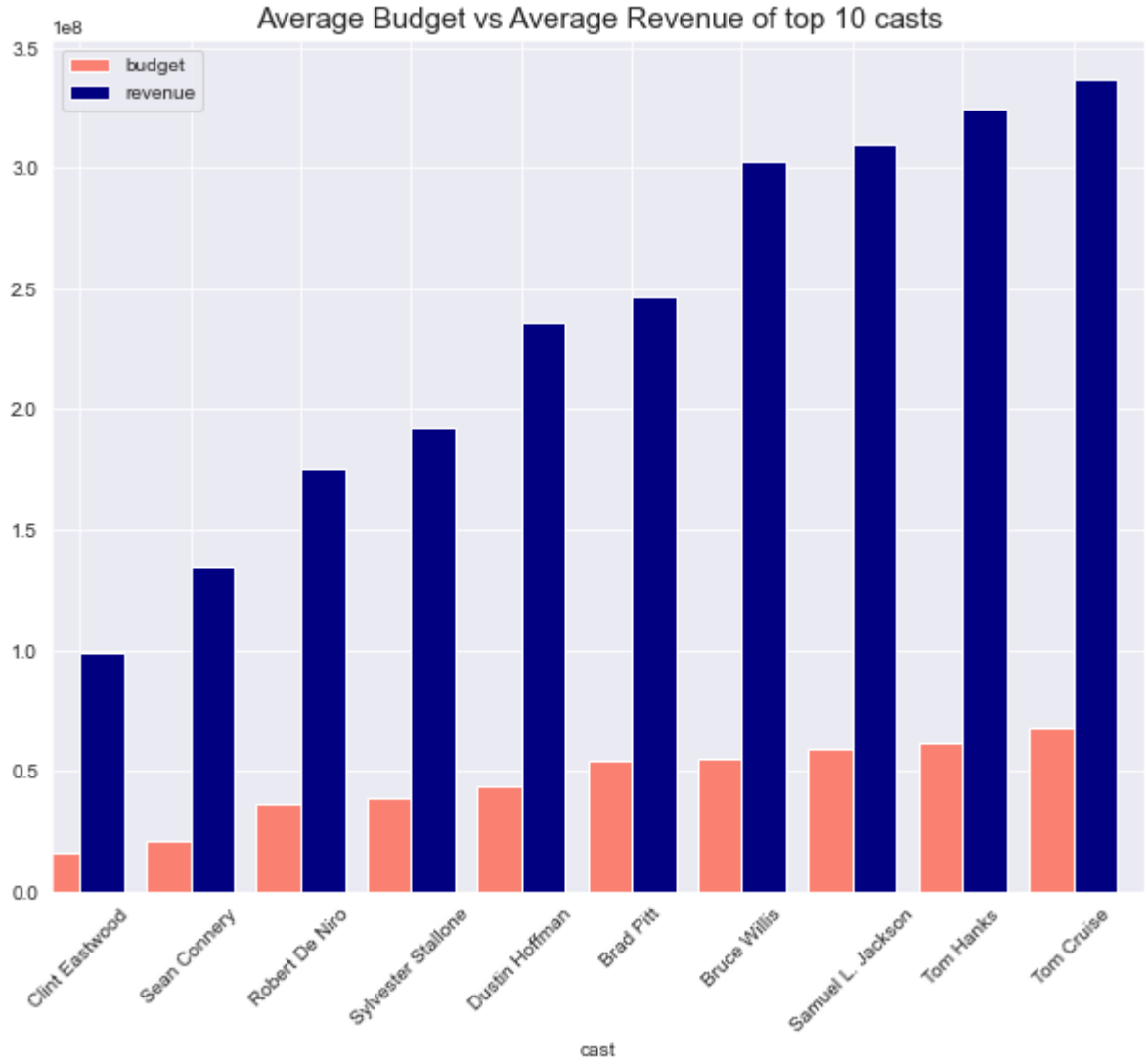
```
#Subset rows in dataframe where top 10 most frequent casts
ten_most_demend_cast = cast_df[cast_df['cast'].isin(['Tom Hanks',
 'Tom Cruise',
 'Robert De Niro',
 'Samuel L. Jackson',
 'Sylvester Stallone',
 'Clint Eastwood',
 'Brad Pitt',
 'Dustin Hoffman',
 'Sean Connery',
 'Bruce Willis'])]

#Make a double bar chart to show the average budget vs average revenue of each cast the
avg_revenue = ten_most_demend_cast.groupby(['cast'])['revenue'].mean().sort_values(asce
avg_budget = ten_most_demend_cast.groupby(['cast'])['budget'].mean().sort_values(ascend

fig = plt.figure(figsize=(10,8))
axes = fig.add_subplot(111)
width = 0.4

avg_budget.plot(kind='bar', color='salmon', ax=axes, width=width, position=1)
avg_revenue.plot(kind='bar', color='navy', ax=axes, width=width, position=0)

axes.set_title("Average Budget vs Average Revenue of top 10 casts", size=15)
plt.xticks(rotation = 45)
plt.legend();
```



- Although Tom Hanks is the most in demand actor by showing in 21 movies, he just placed on the 2nd in term of average revenue/budget movies where he is. The top super expensive star is Tom Cruise whose movies are on top of budget and revenue. Clint Eastwood is also contributed to the top 10 movie stars. Interestingly, in this list has no actress.

Q9. How do the vote average score and vote count relate to the revenue of the movies?

In [ ]:
```
#Find the descriptive statistics of vote_average score of the dataset
df_commercial_success['vote_average'].describe()
```

Out[ ]:
```
count    1405.000000
mean        6.477936
std         0.745985
min         3.400000
25%         6.000000
50%         6.500000
75%         7.000000
max         8.400000
Name: vote_average, dtype: float64
```

- To be better analyzing, we divide the vote average score into 3 groups based on the quartiles. Low_Score: 3.4-6, Medium_Score: 6-7, High_Score: 7-8.4

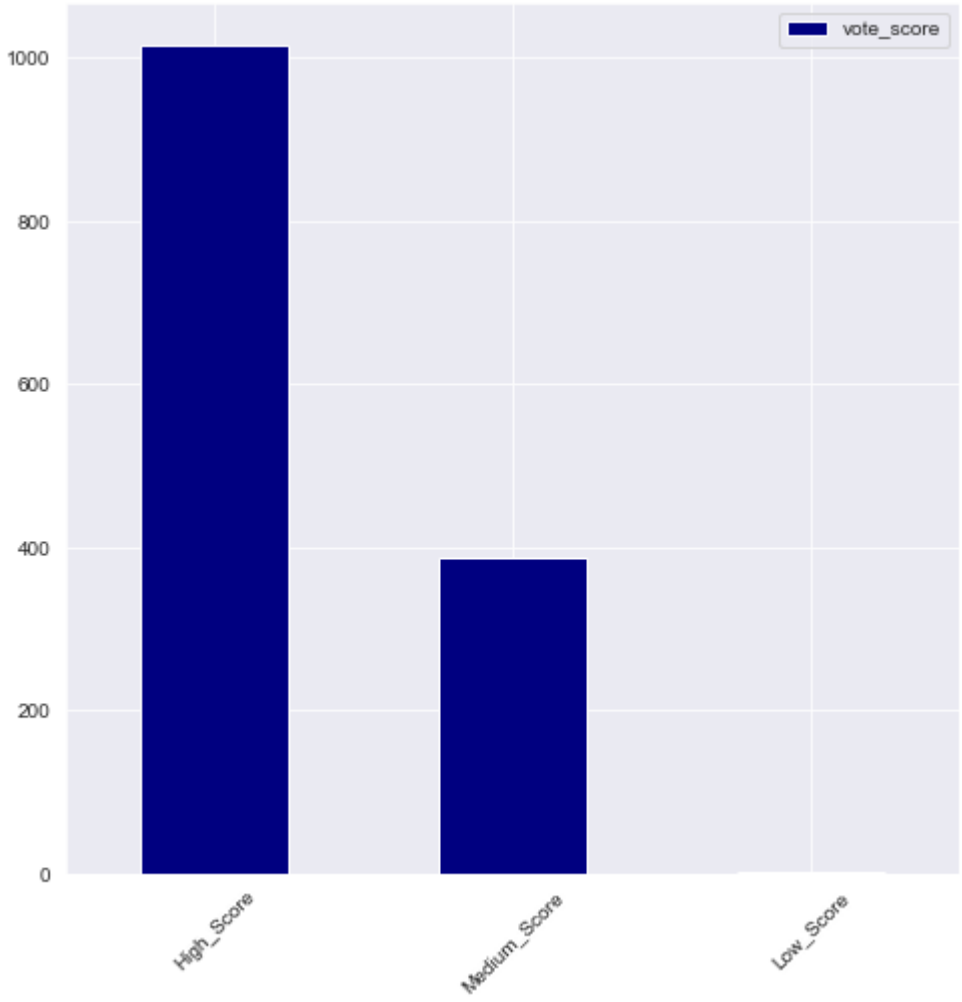
In [ ]:
```
# Import warnings to ignore any warning after running the 'cut' method.
import warnings
warnings.filterwarnings('ignore')

# Divide the vote score into 3 labels and make a nem column named "vote_score" to cater
vote_labels = ['Low_Score', 'Medium_Score', 'High_Score']
vote_bins = [0, 3.4, 6, 8.4]
df_commercial_success['vote_score'] = pd.cut(df_commercial_success['vote_average'], bin
```

In [ ]:
```
# How many movies got Medium to High Score
df_commercial_success['vote_score'].value_counts()
```

Out[ ]:
```
High_Score      1017
Medium_Score     387
Low_Score          1
Name: vote_score, dtype: int64
```

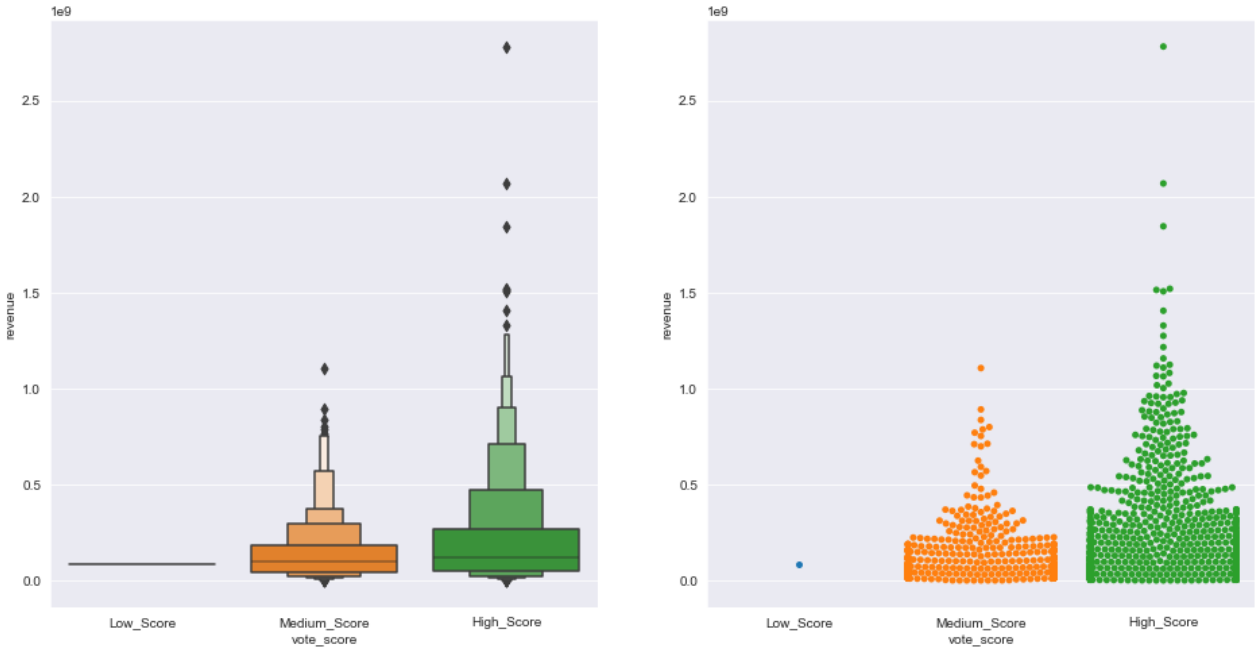
In [ ]:
```
# Plot the bar chart to see how many moives are in each group
fig, axes = plt.subplots(figsize=(8, 8))
df_commercial_success['vote_score'].value_counts().plot(kind='bar', color="navy")
plt.xticks(rotation=45)
plt.legend();
```



- In this commercial succesful dataset, most of the moives got the high score & medium score from people with about 1400 movies.

In [ ]:

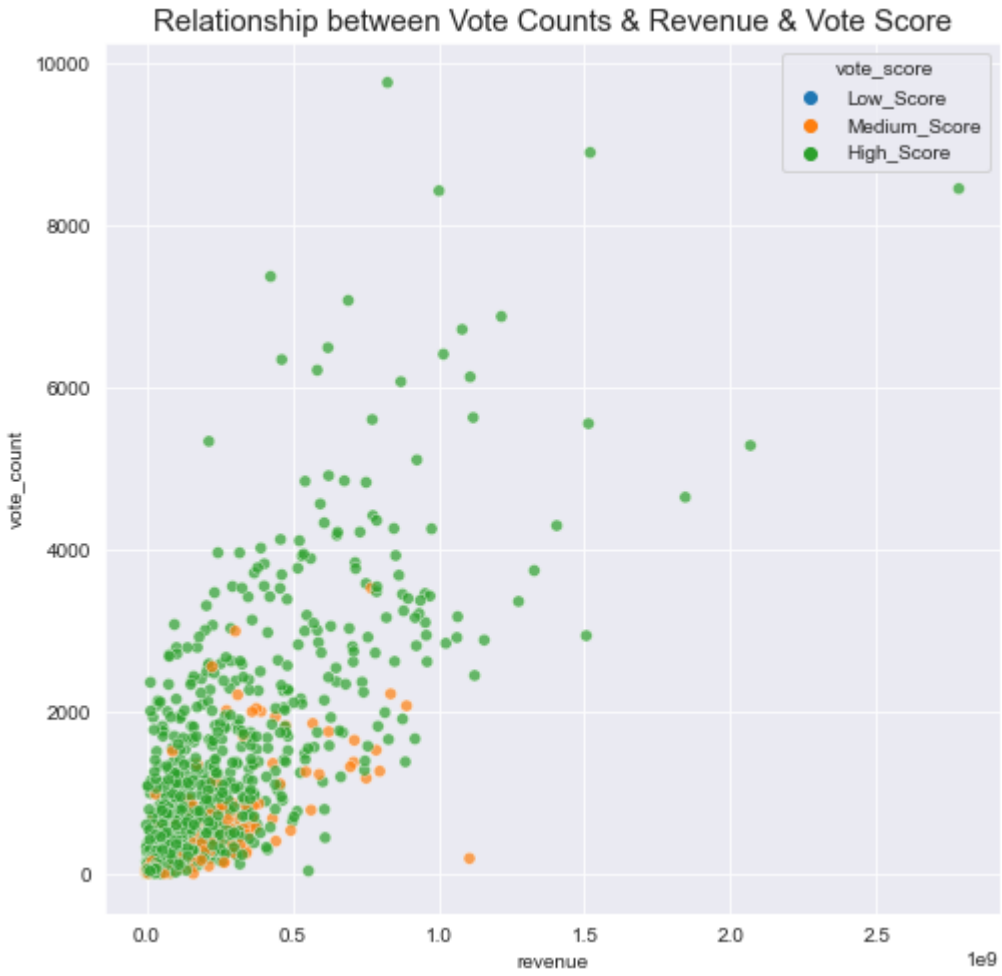
```
# Plot distribution of revenue over the vote score for better understanding how vote sc
fig, axes = plt.subplots(figsize=(16, 8), ncols=2)
sns.boxenplot(y=df_commercial_success['revenue'], x=df_commercial_success['vote_score']
sns.swarmplot(y=df_commercial_success['revenue'], x=df_commercial_success['vote_score']
```



- Most the movie with medium score are below 250M. On the other hand, the revenue of high score movies are largely distributed up to 500M and has more outliers where the revenue can reach up more than 2.5B.

In [ ]:

```
# Plot the scatter plot between vote_count, revenue and vote_score
fig, axes = plt.subplots(figsize=(8, 8))
axes = sns.scatterplot(x='revenue', y='vote_count', data=df_commercial_success, hue='vo
axes.set_title("Relationship between Vote Counts & Revenue & Vote Score", size=15);
```



- The scatter plot show that there is a strong positive relationship between the number of vote count and revenue.

Correlation Matrix

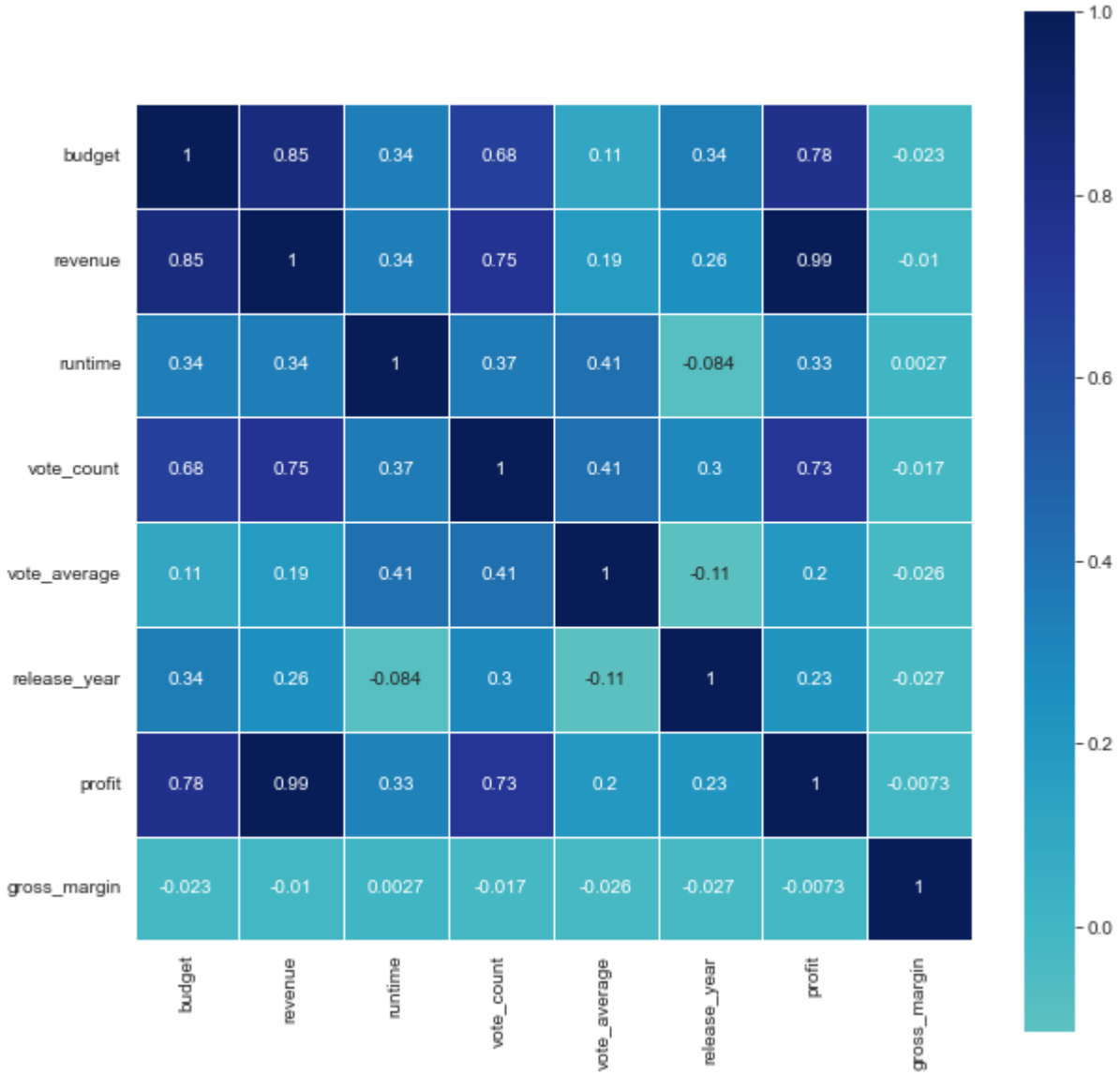
In [ ]:
```
# Calculate the correlation matrix of the successful movies
corr = df_commercial_success.corr()
corr
```

Out[ ]:

	budget	revenue	runtime	vote_count	vote_average	release_year	profit	gross_
budget	1.000000	0.847148	0.339160	0.675210	0.106634	0.343133	0.781129	-0.
revenue	0.847148	1.000000	0.338784	0.748199	0.187906	0.255682	0.993495	-0.
runtime	0.339160	0.338784	1.000000	0.367634	0.411328	-0.084470	0.325402	0.
vote_count	0.675210	0.748199	0.367634	1.000000	0.410295	0.296781	0.734465	-0.
vote_average	0.106634	0.187906	0.411328	0.410295	1.000000	-0.114935	0.197946	-0.
release_year	0.343133	0.255682	-0.084470	0.296781	-0.114935	1.000000	0.226902	-0.
profit	0.781129	0.993495	0.325402	0.734465	0.197946	0.226902	1.000000	-0.
gross_margin	-0.022732	-0.010385	0.002709	-0.016593	-0.026461	-0.026534	-0.007332	1.

In [ ]:
```
fig, axes = plt.subplots(figsize=(10, 10))

axes = sns.heatmap(corr, annot=True, center=0,  linewidths=1, square=True, cmap="YlGnBu
```



- The correlation matrix has shown us that revenue is strongly positive correlated to vote_count & budget. Although the correlation has indicated no strong relation between revenue and vote_average, the distribution of the data as above analysis has shown us the most successful movies got a High Score in average.

Q10. What are words most used in tagline?

In [ ]:
```
# Let's install the packages WordCloud: pip install WordCloud;
# Import necessary packages from world cloud to use
from wordcloud import WordCloud, STOPWORDS, ImageColorGenerator
```

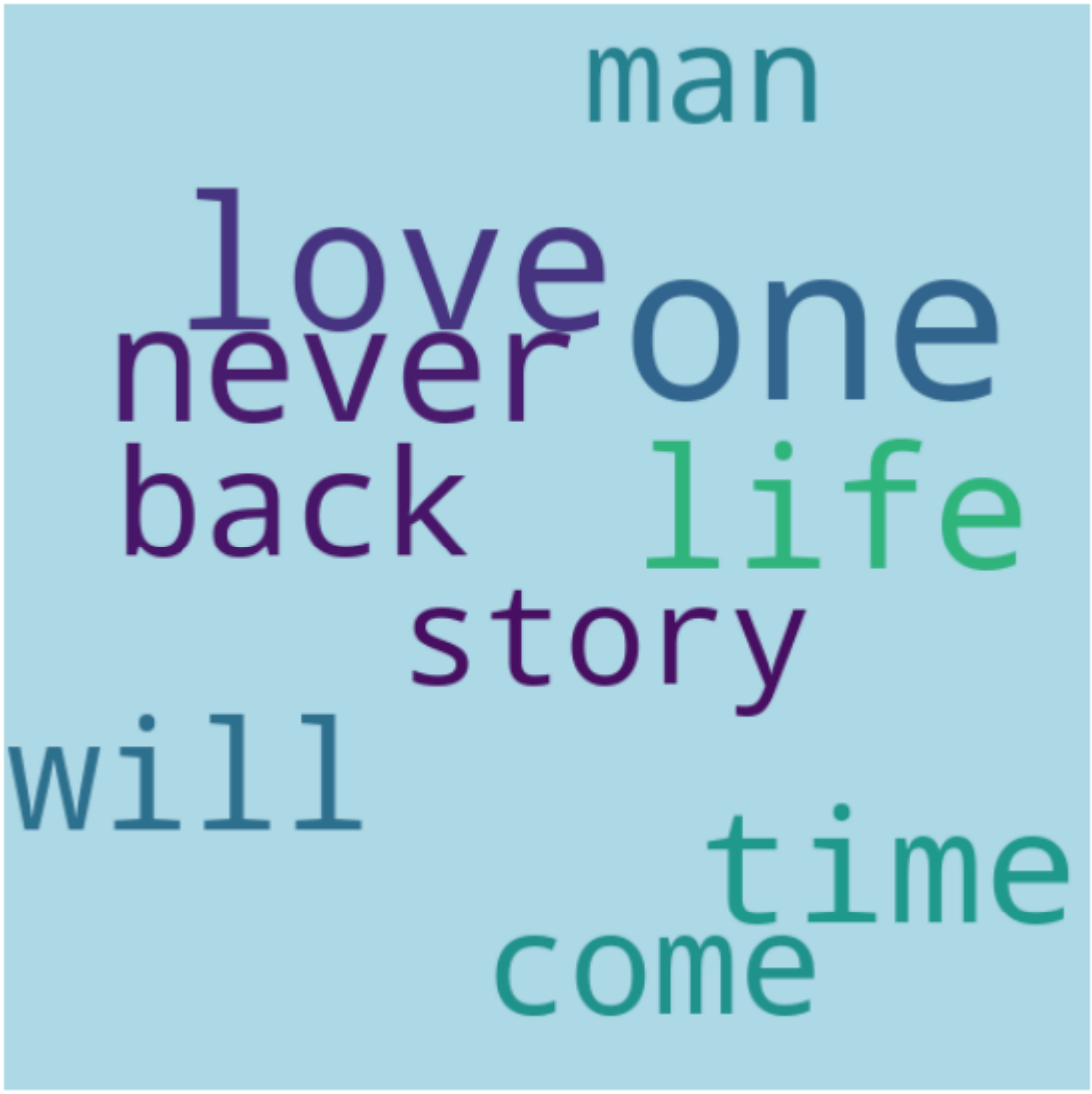
In [ ]:
```
# Drop any null values in the tagline columns due to the world cloud is unable to plot
df_tagline_non_na= df_commercial_success['tagline'].dropna()
df_tagline_non_na.isnull().sum()
```

Out[ ]: 0

In [ ]:
```
# Convert the tagline into string and join them all together
word_string=" ".join(str(x) for x in df_tagline_non_na)
```

In [ ]:
```
# Plot the world cloud to find the top 10 words mostly used to write in the tagline.
plt.figure(figsize=(10,10))
wc = WordCloud(background_color="lightblue", stopwords = STOPWORDS, max_words=10, max_f
wc.generate(word_string)

plt.imshow(wc.recolor( colormap= 'viridis' , random_state=7), interpolation="bilinear")
plt.axis('off');
```



- Would you consider make the next succesful movie with this combination of word for tagline?
- Guest what next succesful movie content is:

> "one man life time story will never come back love"

Limitation:

- To analyze in the aspects of budget & revenue, we need both of data available. However, there are either zero revenue & budget data therefore a large number of data is dropped out.
- There are a couple of movies in which runtime is zero. That's possibly not a right value.
- There are a couple of null data especially on tagline columns where the movies have no hooked sentences. That could be a drawback of this dataset as analyzing the frequent of words used in tagline.
- The cast & genres columns have the pipe (|) separator that needed to be took out to get the list of cast and genres for analysis.

Conclusion

- A successful commercial movie has at least 200% in gross margin where there are more than 1405 one to meet this criteria.
- Clint Eastwood is both one of the best director and cast in movies he is in. Steven Spielberg is the supper rock star when he has made 20 movies with gross margin bigger than 200%. We definitely consider to hire these most talent directors to direct the next blockbuster.
- Avatar has the biggest number of revenue with more than 2.5B. That has not been beat by any movies yet. The runner up is Star Wars: The Force Awakens but it is still far below around 500M compared to super blockbuster Avatar. Interestingly, Director James Cameron owns 2 movies (Avatar & Titanic) in the top 10 revenue movies, but Steven Spielberg is the one who has the highest number of successful one.
- The final cut for a movie is in range from 94-122min. That is where the most successful movies are. That's also a preference for the post-production process.
- Drama, Comedy, Thriller are the top 3 genres have been made, however, they are not commercially profitable compared to Fantasy, Animation and Adventure where they generate a heap of revenue in average.
- Tom Cruise and Tom Hanks are 2 super stars because movies in which those generate a lot of revenue. The top 10 casts have no female. That's interesting point in the film industry.
- A strong positive correlation for the vote count and vote score to the revenue. The more public votes criticize positively, the more revenue that the movie have.
- The final word cloud in tag line give us the sense of which words in the hooked line in order to catch the attention for the movie which is "one man life time story will never come back love".

Reference

- https://stackoverflow.com/questions/10880813/typeerror-sequence-item-0-expected-string-int-found
- https://www.kaggle.com/code/olgaberezovsky/word-cloud-using-python-pandas/notebook